



Yakeen Plus (2025)

Practice Test - 15

DURATION :200 Minutes

DATE : 16/12/2024

M.MARKS :720

ANSWER KEY

PHYSICS

1. (1)
2. (4)
3. (2)
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199. (1)
200. (3)

SECTION-(I) PHYSICS

1. (1)

The orbital velocity of spaceship in circular orbit,

$$v_0 = \sqrt{GM} / r$$

If spaceship is very close to earth surface,

r = radius of earth = R

$$\therefore v_0 = \sqrt{\frac{GM}{R}} = \sqrt{\frac{R^2 g}{R}} = \sqrt{Rg} \left(\text{since, } g = \frac{GM}{R^2} \right)$$

Here, R = 6400 km = 6.4×10^6 m, g = 9.8 m/s²

$$\therefore \sqrt{6.4 \times 10^6 \times 9.8}$$

$$= 7.92 \times 10^3 \text{ m/s} = 7.92 \text{ km/s}$$

The escape velocity of spaceship,

$$v_e = \sqrt{(2Rg)} = \sqrt{2}v_0 = \sqrt{2} \times 7.92$$

$$= 11.20 \text{ km/s}$$

 $\therefore$ Additional velocity required

$$= 11.20 - 7.92 = 3.28 \text{ km/s}$$

2. (4)

From conservation of energy,

Total energy at surface = total energy at height H.

$$\therefore \frac{1}{2}mv_1^2 - \frac{GM_e M}{R} = -\frac{GM_e M}{R+H}$$

Solving this, we get

$$H = \frac{R}{\frac{2GM_e}{v_1^2} - 1} = \frac{R}{\frac{v_e^2}{v_1^2} - 1} \left(\because v_e = \sqrt{\frac{2GM_e}{R}} \right)$$

$$\text{Here, } v_1 = \frac{1}{3}v_e \Rightarrow \frac{v_e}{v_1} = 3$$

$$\therefore H = \frac{R}{9-1} = \frac{R}{8}$$

3. (2)

From Kepler's third law,

$$\frac{T_1^2}{T_2^2} = \frac{r_1^3}{r_2^3}$$

$$\Rightarrow r_2 = \left(\frac{T_2}{T_1} \right)^{2/3} \times r_1$$

Here, T₁ = 1h, T₂ = 8h and r₁ = 10⁴ km

$$\Rightarrow r_2 = \left(\frac{8h}{1h} \right)^{2/3} \times 10^4 = 4 \times 10^4 \text{ km}$$

The speed of satellite S₂ relative to S₁,

$$|v_2 - v_1| = \frac{2\pi r_2}{T_2} - \frac{2\pi r_1}{T_1} = 2\pi \left[\frac{r_2}{T_2} - \frac{r_1}{T_1} \right]$$

$$= 2\pi \left[\frac{4 \times 10^4}{8} - \frac{10^4}{1} \right] \text{ km/h}$$

$$= 2\pi \times \frac{1}{2} \times 10^4 = \pi \times 10^4 \text{ km/h}$$

4. (3)

As we know, Young's modulus of elasticity,

$$Y = \frac{\text{Stress}}{\text{Strain}} = \frac{\text{Force / Area}}{\frac{\text{Change in length}}{\text{Original length}}}$$

In first case,

$$\Rightarrow Y = \frac{T_1 \times L}{A \times (l_1 - L)} \quad \dots\dots(i)$$

Where, L is the original length of the wire and A is cross-section area of the wire.

In second case,

$$Y = \frac{T_2 \times L}{A \times (l_1 - L)} \quad \dots\dots(ii)$$

From Eqs. (i) and (ii), we get

$$L = \frac{T_2 l_1 - T_1 l_2}{T_2 - T_1}$$

5. (3)

As we know that, the Young's modulus,

$$Y = \frac{\text{Stress}}{\text{Strain}} = \frac{F / A}{\Delta l / L}$$

$$\text{In first case, } Y = \frac{F / A_1}{l_1 / L} \Rightarrow F = \frac{Y \cdot l_1}{L} A_1 \quad \dots\dots(i)$$

Similarly for second case,

$$F = \frac{Y l_2}{2L} A_2 \quad \dots\dots(ii)$$

From Eqs. (i) and (ii), we get

$$\frac{Y l_1}{L} A_1 = \frac{Y l_2}{2L} A_2$$

$$\Rightarrow l_1 A_1 = \frac{l_2}{2} A_2$$

$$\Rightarrow l_1 \pi r^2 = \frac{l_2}{2} \cdot \pi (2r)^2$$

6. (1)

By Bernoulli's theorem,

$$p_1 + \frac{1}{2} \rho v_1^2 = p_2 + \frac{1}{2} \rho v_2^2 \quad (\text{for horizontal tube})$$

$$p_1 - p_2 = \frac{1}{2} \rho (v_2^2 - v_1^2) \quad \dots\dots(i)$$

But, by the principle of continuity,

$$Q = A_1 v_1 = A_2 v_2 \quad \dots\dots(ii)$$

From Eqs. (i) and (ii),

$$p_1 - p_2 = \frac{1}{2} \rho \left(\frac{Q^2}{A_2^2} - \frac{Q^2}{A_1^2} \right)$$

$$\Rightarrow Q = A_1 A_2 \sqrt{\frac{2(p_1 - p_2)}{\rho(A_1^2 - A_2^2)}}$$

$$= A_1 A_2 \sqrt{\frac{2\Delta p}{\rho(A_1^2 - A_2^2)}}$$

(4)

(3)

(2)

(4)

(3)

(4)

$$\frac{E_1}{E_2} = \left(\frac{600}{200} \right)^4 \Rightarrow \frac{E_1}{E_2} = \frac{81}{1}$$

13. (1)

Given, $t_1 = 20^\circ \text{C}$ and $t_2 = 100^\circ \text{C}$
Let the value of the body = V , then

$$v_1 = \frac{2}{3}V \text{ and } v_2 = \frac{3}{4}V$$

As we know that,

$$V_2 = V_1(1 + \gamma \Delta t)$$

$$\text{Thus, } \gamma = \frac{V_2 - V_1}{V_1 \Delta t} = \frac{V_2 - V_1}{V_1(t_2 - t_1)}$$

$$\text{Putting the value, } \frac{\left(\frac{3}{4} - \frac{2}{3}\right)V}{\frac{2}{3}V(100 - 20)} = \frac{1/12}{2/3 \times 80}$$

$$= 15.625 \times 10^{-4} / ^\circ \text{C}$$

14. (3)

Given $t_1 = 12^\circ \text{C}$ and $t_2 = 100^\circ \text{C}$
The given situation can be shown as.

A		B	
12°C	K_1	x	K_2
			0°C

Rate of flow of heat will be equal in both the slabs, because they are in series combination.

$$\therefore (12 - x)K_1 = K_2(x - 0)$$

$$(12 - x)K_2 = 2k_2(x - 0)$$

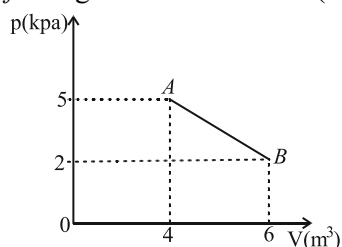
$$12 - x = 2x \Rightarrow = 4^\circ \text{C} \left(\therefore K_1 = \frac{k_2}{2} \right)$$

The temperature difference across slab,
 $A = (12 - x) = (12 - 4) = 8^\circ \text{C}$

15. (3)

From graph, $p_1 = 5 \text{ kPa}$, $p_2 = 2 \text{ kPa}$,
 $V_1 = 4 \text{ m}^3$ and $V_2 = 6 \text{ m}^3$

f = degree of freedom = 5 (for diatomic)



The change in internal energy,

$$\Delta U = \frac{f}{2}(p_2 V_2 - p_1 V_1) = \frac{5}{2}(12 - 20) = -20 \text{ kJ}$$

16. (3)

From ideal gas equation,

$$pV = nRT \quad \dots\dots(i)$$

where, symbols have their usual meaning.

For isobaric process (in which pressure remains constant),

$$\text{amount of heat, } Q = nC_p \Delta T \quad \dots\dots(ii)$$

$$\text{Work done, } W = p \Delta V \quad \dots\dots(ii)$$

From Eqs. (i), (ii) and (iii), we get

$$\frac{Q}{W} = \frac{nC_p \Delta T}{nR \Delta T} = \frac{C_p}{R} = \frac{C_p}{C_p - C_V} \quad [\because R = C_p - C_V]$$

$$= \frac{C_p / C_V}{\frac{C_p}{C_V} - 1} = \frac{\gamma}{\gamma - 1} \quad \left(\because \gamma = \frac{C_p}{C_V} \right)$$

17. (2)

In the process,

$$dQ = 8 \times 10^5$$

$$dW = 6.5 \times 10^5 \text{ J}$$

From first law of thermodynamics,

$$dU = dQ - dW$$

$$dU = 8 \times 10^5 - 6.5 \times 10^5$$

$$= 1.5 \times 10^5 \text{ J}$$

In the second process, change in internal energy dU remains the same.

$$\therefore dU = dQ - dW$$

$$= 1 \times 10^5 - 1.5 \times 10^5$$

$$dW = -0.5 \times 10^5 \text{ J}$$

Negative sign shows that work is done on the gas.

18. (2)

From first law of thermodynamics, we have

$$\Delta U = Q - W$$

Amount of heat = Mass $\times$ Latent heat

$$\therefore \text{We have, } Q = mL = 1 \times 2240$$

$$W = 168 \text{ J (given)}$$

$$\therefore \Delta U = 2240 - 168 = 2072 \text{ J}$$

19. (3)

In the given figure, spring with spring constants k_1 and k_2 are in parallel and that are with k_3 and k_4 are also in parallel. The two combinations are in series with each other.

$$\therefore k_1 = k_2 = k$$

$$\text{Also, } k_3 = k_4 = k$$

$$\Rightarrow k_1 + k_2 = k + k = 2k$$

$$\text{and } k_3 + k_4 = k + k = 2k$$

$$\frac{1}{k_{eq}} = \frac{1}{2k} + \frac{1}{2k}$$

$$k_{eq} = k$$

$$\text{Time period is given by, } T = 2\pi \sqrt{\frac{m}{k}}$$

Where, m is mass and k is spring constant.

20. (2)
Root mean square (*rms*) speed of gas molecules does not depend on the pressure of gas (if temperature remains constant) because pressure (p) $\propto$ density (ρ) (Boyle's law).
If pressure is increased n times, density will also increase by n times, but v_{rms} remains constant.

21. (2)
For mixture of gases,

$$C_V = \frac{\frac{3}{2}R + \frac{5}{2}R}{1+1} = 2R \left[\begin{array}{l} \because \text{For O } V_V = \frac{5}{2}R \\ \text{and for He, } C_V = \frac{3}{2}R \end{array} \right]$$

$$C_p = 2R + R = 3R \quad \left[\because C_p = C_V + R \right]$$

$$\therefore \frac{C_p}{C_V} = \frac{3R}{2R} = 1.5$$

22. (2)
Van der Waal's gas equation for n mole of real gas,

$$\left(p + \frac{n^2 a}{V^2} \right) (V - nb) = nRT$$

(where a and b are der Waal's constants)

$$\text{or } p = \left(\frac{nRT}{V - nb} \right) - \frac{n^2 a}{V^2} \quad \dots\dots\dots(i)$$

Given equation,

$$p = \left(\frac{RT}{2V - b} - \frac{a}{4V^2} \right) = \left[\left(\frac{\frac{1}{2}RT}{V - b/2} \right) - \left(\frac{1}{2} \right)^2 \cdot \frac{a}{V^2} \right] \quad \dots\dots\dots(ii)$$

On comparing Eqs. (i) and (ii), we get

$$n = \frac{1}{2}$$

$$\text{We know that, } n = \frac{m}{M}$$

[where, m = mass of gas and M = molecular mass of gas]

$$\text{or } m = nM = \frac{1}{2} \times 44 = 22g$$

$$[M_{CO_2} = 12 + 2 \times 16 = 44g]$$

23. (4)
From Boyle's law,
when T is constant, $pV = \text{constant}$ $\left(\text{or } P \propto \frac{1}{V} \right)$.
When volume is decreased by 10%
i.e. Volume becomes $\frac{90}{100}$, the pressure must become $100/90$
 $\therefore$ % increase in pressure = $\frac{(100 - 90) \times 100}{90}$
= 11.1%

24. (3)
(i) The dotted line in the diagram shows that there is not deviation in the value of $\frac{pV}{nT}$ for different temperatures T_1 and T_2 for increasing pressure, So, this gas behaves ideally. Hence, dotted line corresponds to 'ideal' gas behaviour.

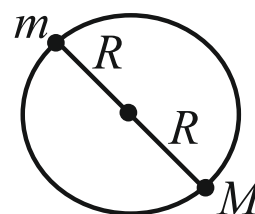
(ii) At high temperature, the deviation of the gas should be less and at low temperature, the deviation of gas should be more. In the graph, deviation for T_2 is greater than for T_1 Thus, $T_1 > T_2$.

(iii) Since, the two curves intersect at dotted line. So, the value $\frac{pV}{nT}$ at that point on the Y-axis is same for all gases.

25. (3)
Gravitational force of attraction between the particles.

$$F = \frac{Gm^2}{4R^2}$$

$$F = mR\omega^2 = \frac{mv^2}{R}$$



$$\therefore \frac{Gm^2}{4R^2} = \frac{mv^2}{R}$$

$$v = \sqrt{\frac{GM}{4R}} = \frac{1}{2} \sqrt{\frac{GM}{R}}$$

26. (2)
 $F = \frac{GM}{r^2} \Rightarrow F \propto r^{-2}$

But is $F \propto r^{-5/2}$

Potential energy for a conservative force is defined as:

$$U = - \int F \cdot dr$$

$$\int r^{-5/2} dr = r^{-3/2}$$

$$\therefore U \propto r^{-3/2}$$

27. (3)

Equation of SHM is given by

$$y_1 = 0.1 \sin \left(100\pi t + \frac{\pi}{3} \right)$$

$$v_1 = \frac{dy_1}{dt} = 0.1 \times 100\pi \cos \left(100\pi t + \frac{\pi}{3} \right)$$

$$= 10\pi \cos \left(100\pi t + \frac{\pi}{3} \right)$$

$$= 10\pi \sin \left(\frac{\pi}{2} + 100\pi t + \frac{\pi}{3} \right)$$

Also, $y_2 = 0.1 \cos \pi t$

$$v_2 = \frac{dy_2}{dt} = -0.1\pi \sin \pi t$$

$$= 0.1\pi \sin (\pi + \pi t)$$

At $t = 0$, phase difference of v_1 w.r.t. v_2

$$= \left(\frac{\pi}{2} + \frac{\pi}{3} \right) - \pi = -\frac{\pi}{6}$$

28. (2)

Given, $v \frac{dv}{dx} = \omega^2 x$

On integrating within the limit $\int_{v_0}^v v dv = \int_0^x -\omega^2 x dx$

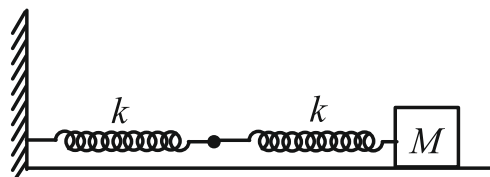
$$\Rightarrow \left[\frac{v^2}{2} \right]_{v_0}^v = -\omega^2 \left[\frac{x^2}{2} \right]_0^x \Rightarrow v^2 - v_0^2 = -\omega^2 x^2$$

$$\Rightarrow v = \sqrt{v_0^2 - \omega^2 x^2}$$

29. (2)

Let when the oscillating mass is at a distance x towards right from its equilibrium position, the instantaneous extensions in the springs of force constant k , is

$$x = x_1 = x_2$$



Since, the springs are in series the restoring force exerted by each spring on mass m is same.

Then,

$$F = -kx_1 = -kx_2$$

$$\text{and } x = x_1 + x_2 = -F \left(\frac{1}{k} + \frac{1}{k} \right) = \frac{-F}{k}$$

$$\therefore F = -\frac{k}{2} x$$

$$\Rightarrow \text{Effective force constant is } \frac{k}{2}.$$

Hence, frequency of oscillation is

$$n = \frac{1}{2\pi} \sqrt{\frac{k}{2M}}$$

30. (4)

The time period of a simple pendulum of a length l is given by

$$T = 2\pi \sqrt{\frac{l}{g}} \quad \dots(i)$$

When length of pendulum increases by 21% we have, total increases length of pendulum.

$$l' = 1.21 l$$

$$\therefore T' = 2\pi \sqrt{\frac{1.21l}{g}} \quad \dots(ii)$$

Dividing Eq. (i) by Eq. (ii), we get

$$\frac{T}{T'} = \sqrt{\frac{1}{1.21}}$$

$$\frac{T}{T'} = \sqrt{\frac{121}{100}} T = 1.1T$$

Percent increase in time period

$$= \frac{1.1T - T}{T} \times 100 = \frac{0.1T}{T} \times 100 = 10\%$$

Alternative Time period of simple pendulum

$$T = 2\pi \sqrt{\frac{l}{g}} \Rightarrow T \propto \sqrt{l}$$

$$\therefore \frac{\Delta T}{T} = \frac{1}{2} \cdot \frac{\Delta l}{l}$$

$$\text{Since, } \frac{\Delta l}{l} = 21\%$$

Increase in the time period of pendulum is

$$\frac{\Delta T}{T} = \frac{1}{2} \times 21\% \approx 10\%$$

31. (3)

$$\text{The frequency of string is } n = \frac{1}{2l} \sqrt{\frac{T}{m_l}}$$

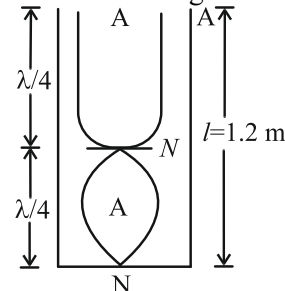
where, T is the tension, m_l is the mass per unit length and l is the length of string,

$$\text{So, } m_l = \frac{10 \times 10^{-3}}{1} = 10^{-2} \text{ kg-m}^{-1}$$

$$\Rightarrow 50 = \frac{1}{2} \sqrt{\frac{T}{10^{-2}}} \Rightarrow T = \frac{2500 \times 4}{100} = 100 \text{ N}$$

32. (1)

Let l be the length of air column.



In first overtone mode,

$$l = \frac{3\lambda}{4} \text{ (where, } \lambda \text{ is wavelength)}$$

$$\therefore \frac{\lambda}{4} = \frac{l}{3} = \frac{1.2}{3} = 0.4 \text{ m}$$

Pressure variation will be maximum at displacement nodes, i.e., at 0.4 m from the open end.

33. (1)

There are four beats between P and Q therefore, the possible frequencies of P are 246 or 254 (i.e. 250 ± 4) Hz.

When the prong of P is filed, its frequency becomes greater than the original frequency.

The beats between P and Q will be more than 4. But it is given that the beats are reduced to 2, therefore, 254 is not possible.

Therefore, the required frequency must be 246 Hz. (This is true, because on filing the frequency may increase to 248, given 2 beats with Q of frequency 250 Hz).

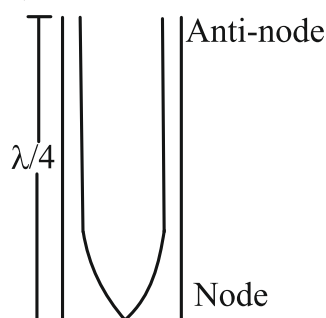
34. (3)

A closed pipe produces only odd harmonics. The frequency of note emitted from the pipe for v being velocity of sound in air, is

$$f = n \left(\frac{v}{4l} \right)$$

and l is length of pipe. $f = n \times \text{fundamental frequency}$

$$\left(\because \text{fundamental frequency} = \frac{v}{4l} \right)$$



$\therefore$ we know that human ear can hear frequencies upto 20,000 Hz, hence

$$20,000 = n \times 15000$$

$$\Rightarrow n = \frac{20,000}{1500} \approx 13$$

Maximum possible harmonics obtained are

1, 3, 5, 7, 9, 11, 13.

Hence, man can hear upto 13th harmonic

$$= 7 - 1 = 6$$

Note: First overtone of the closed pipe is called the third harmonic.

35. (2)

$$f = (2n + 1) \frac{v}{4l}$$

$$f = (2n + 1) \frac{v}{4l}$$

$$f = (2n + 1) \times 80$$

$$n = 0, 1, 2, 3, \dots$$

$$f = (2 \times 0 + 1) 80$$

$$= 80 \text{ Hz}$$

$$f = (2 \times 1 + 1) \times 80$$

$$= 80 \text{ Hz}$$

$$f = (2 \times 1 + 1) \times 80$$

$$= 240 \text{ Hz}$$

$$f = (2 \times 2 + 1) \times 80$$

$$= 400 \text{ Hz}$$

36. (1)

$$y = 8 \sin^2 \pi \left[\frac{t}{0.02} - \frac{x}{100} \right]$$

$$y = 4 - 4 \cos 2\pi \left[\frac{t}{0.02} - \frac{x}{100} \right]$$

$$\text{Amplitude} = 4$$

$$\omega = \frac{2\pi}{0.02} = 100\pi$$

$$2\pi f = 100\pi$$

$$f = 50 \text{ Hz}$$

$$v = \frac{W}{k} = \frac{100\pi}{\frac{2\pi}{100}} = 50,000 \text{ cm/s}$$

$$V = 50 \text{ m/s}$$

$$V_{\text{max}} = A\omega = 4 \times 100\pi = 400\pi$$

37. (3)

$$Y = 0.02 \cos \left(50\pi t + \frac{\pi}{2} \right) \cos 10\pi x$$

$$Y = 2A \cos \omega t \cos kx$$

$$2A = 0.02$$

$$A = 0.01$$

$$K = \frac{2\pi}{\lambda} = 10\pi$$

$$\lambda = \frac{1}{5} = 0.2 \text{ m}$$

$$v = \frac{w}{k} = \frac{50\pi}{10\pi}$$

$$= 5 \text{ m/s}$$

Nodes will be formed where displacement of particle is zero.

$$\cos 10\pi x = 0$$

$$10\pi x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}$$

$$x = 0.5, 0.15, 0.25$$

$$\cos 10\pi x = 1$$

$$x = 0.1, 0.2, 0.3$$

38. (3)

$$v = \sqrt{\frac{3RT}{M}}$$

$$v = \sqrt{\frac{3P}{s}}$$

39. (4)

$$T = 2\pi \sqrt{\frac{1}{g_{\text{eff}}}}$$

$g_{\text{eff}} = g$, When lift at rest

$g_{\text{eff}} = g$ $v = \text{const}^n$

$g_{\text{eff}} = (g + a)$, When lift moving upward with acceleration (a)

40. (2)

$$\delta = \frac{PM}{RT}$$

41. (3)

$$C = \frac{Q}{m\Delta t}$$

For adiabatic $Q = 0$, $C = 0$

for isothermal $\Delta t = \infty$ $C = \infty$

$C = M_w \times \text{specific heat capacity}$.

42. (4)

For process A, $V = \text{constant}$

$W = \text{constant}$

$W = 0$

$P \downarrow$ $T \downarrow$

$du = nC_v dt$

$\therefore$

$du = -ve$

For process B-

$W = +ve$

$du = nC_v dt$

$= -ve$

for process C

$T = \text{constant}$

$$W = nRT \log_e \frac{V_2}{V_1}$$

$W = +ve$

$d\phi = dw = +ve$

$du = 0$

For process D

$P = \text{const}^n$

$V \uparrow$ $T \uparrow$ $W = pDv$

$W = +ve$

$dQ = du + dw = +ve$

43. (1)

A good cooking utensil should have low specific heat, low thermal capacity, and high thermal conductivity.

44. (4)

$A \rightarrow B$, Volume of substance decreases.

$B \rightarrow C$, Volume of substance decreases.

$\Delta\phi = ms \Delta T$

$$\frac{\Delta Q}{\Delta T} = ms$$

$$Q = mL$$

Latent heat of fusion of substance is independent of portion AB of graph.

45. (2)

Boiling point	212 F	100°C	373.15 K
	Fahrenheit	Celsius	Kelvin
Freezing point	32 F	0°C	273.15 K

46. (3)

$$P = \sigma e A (T^4 - T_0^4)$$

$$P \propto A$$

$$P \propto r^2$$

$$\frac{msdt}{dt} = e\sigma A (T^4 - T_0^4)$$

$$R \propto \frac{1}{r}$$

47. (1)

Temperature- Degree of hotness or coldness of a substance. Internal energy $dU = nC_v dt$

Heat- Heat is an energy in transit means that heat refers specifically to the energy that is being transferred between system as a result of a temp. Difference, rather than energy that is stored with in a system.

$$\text{specific heat} = s = \frac{Q}{m\Delta t}$$

48. (1)

$$\lambda_m T = \text{const}^n$$

49. (4)

Speed of sound

$$V_s = \sqrt{\frac{\gamma RT}{M_w}}$$

rms speed of gas

$$v_{rms} = \sqrt{\frac{3RT}{M_w}}$$

$$\text{Average speed of gas molecules} = \sqrt{\frac{8RT}{\pi M_w}}$$

Most probable speed of gas molecules =

$$V_{MP} = \sqrt{\frac{2RT}{M_w}}$$

50. (1)

Adiabatic Bulk modulus $= \gamma P$

$$\frac{dP}{dV} = -\frac{P}{V} \quad (\text{for isothermal process})$$

$$\gamma = 1 + \frac{2}{f}$$

$$\gamma - 1 = \frac{2}{f}$$

$$\frac{\gamma - 1}{2} = \frac{1}{f}$$

$$f = \frac{2}{\gamma - 1}$$

$$C_p = \frac{\gamma R}{\gamma - 1}$$

$$\frac{C_p}{R} = \frac{\gamma}{\gamma - 1}$$

SECTION-(II) CHEMISTRY

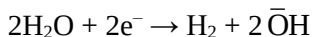
51. (4)

At anode



$$\text{No. of equivalents of } \text{O}_2 = n_{\text{O}_2} \times 4 = \frac{2.4}{22.4} \times 4$$

At Cathode



$$\Rightarrow \text{No. of equivalents of } \text{H}_2 = \frac{2.4 \times 4}{22.4}$$

$$\Rightarrow \text{No. of moles of } \text{H}_2 = \frac{2.4 \times 2}{22.4} \Rightarrow 0.2143$$

$$\Rightarrow \text{Volume of } \text{H}_2 \text{ liberated at STP} = 4.8 \text{ L}$$

52. (3)

$$\Delta G^\circ = -nFE^\circ_{\text{cell}}$$

$$E^\circ_{\text{cell}} = -ve \text{ then } \Delta G^\circ = +ve \text{ i.e.; } \Delta G^\circ > 0$$

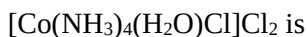
$$\Delta G^\circ = -nRT \log K_{\text{eq}}$$

$$\Delta G^\circ = +ve, K_{\text{eq}} = -ve \text{ i.e., } K_{\text{eq}} < 1$$

53. (4)

For IUPAC naming of complexes, ligands are written in alphabetical order followed by metal ion in its respective oxidation state. Primary valencies are shown outside the coordination sphere.

The correct IUPAC name of



Tetraammineaquachloridocobalt (III) chloride.

54. (1)

The total charge can be calculated using Faraday's constant, which is approximately 96500 coulombs per mole of electrons:

$$Q = \text{Number of moles of electrons} \times \text{Faraday's constant} = 0.01 \text{ moles} \times 96500 \text{C/mole} = 965 \text{C}$$

Current can be calculated using the formula:

$$I = \frac{Q}{t}$$

where Q is the total charge and t is the time in seconds.

Substituting the values:

$$I = \frac{965 \text{ C}}{965 \text{ s}} = 1 \text{ A}$$

55. (3)

$$R = 100 \text{ ohm}, G = \frac{1}{R}$$

$$G^* = 1 \text{ cm}^{-1}$$

$$\kappa = ?$$

$$\kappa = GG^* = \frac{G^*}{R} = \frac{1}{100} \times 1 = 10^{-2} \text{ ohm}^{-1} \text{ cm}^{-1}$$

56. (3)

$$E^\circ = 0.295 \text{ V}$$

$$n = 2$$

$$K = ?$$

$$E^\circ = \frac{0.059}{n} \log K$$

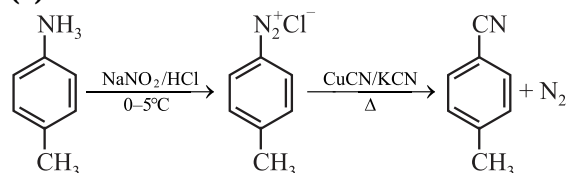
$$\log K = \frac{nE^\circ}{0.059} = \frac{2 \times 0.295}{0.059} = 10$$

$$K = 10^{10}$$

57. (3)

Bromoform	Tribromomethane
Vinyl chloride	Chloroethene
Benzyl chloride	Chlorophenylmethane
Chloroform	Trichloromethane

58. (3)

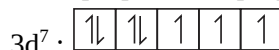


59. (1)

- Reaction of alcohol with SOCl_2 gives alkyl chloride, sulphur dioxide and hydrogen chloride.
- In $\text{S}_{\text{N}}1$ reaction, the order of reactivity of alcohol is $3^\circ > 2^\circ > 1^\circ$.

60. (3)

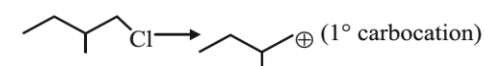
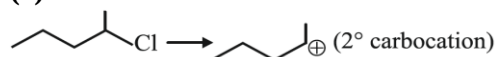
$$\text{Fe}^+ : [\text{Ar}]3\text{d}^6 4\text{s}^1 : [\text{Ar}] 3\text{d}^7 4\text{s}^0$$



: three unpaired electrons,

: hence paramagnetic

61. (3)



In $\text{S}_{\text{N}}1$ mechanism, carbocation is formed as an intermediate.

Stability of carbocation follows the order as: Benzylic $> 3^\circ > 2^\circ > 1^\circ$.

62. (1)

The components of a lead storage battery are a lead anode and a grid of lead packed with lead oxide (PbO_2) as the cathode. The electrolyte is a 38% solution of H_2SO_4 .

77. (3)

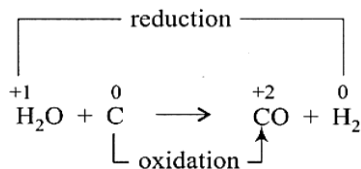
$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{RT}{nF} \ln Q_c$$

at equilibrium $Q_c = K_c$; $E_{\text{cell}} = 0$

$$0 = E_{\text{cell}}^{\circ} - \frac{RT}{nF} \ln K_c$$

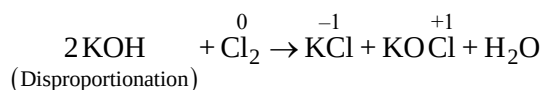
$$E_{\text{cell}}^{\circ} = \frac{RT}{nF} \ln K_c$$

78. (2)



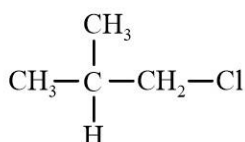
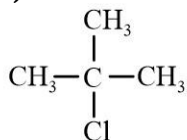
$\therefore$ H_2O is the oxidizing agent. C is the reducing agent.

79. (3)

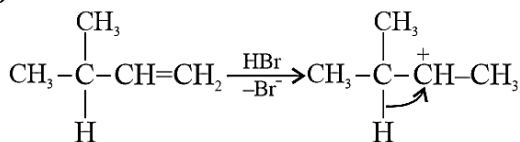


Disproportionation reaction is a redox reaction in which one compound of intermediate oxidation state converts to two compounds, one of higher and one of lower oxidation state.

80. (1)

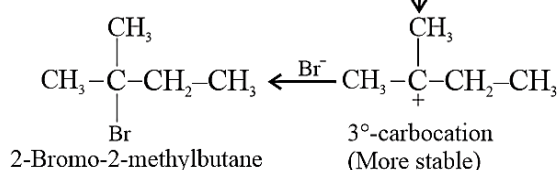


81. (4)

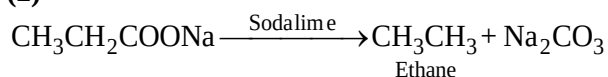


2°-carbocation

1,2-Hydride shift



82. (2)



83. (3)



$$+ 2 + 2x = 0 \therefore x = -1$$

84. (3)

The melting point of neopentane is greater than that of n-pentane but the boiling point of n-pentane is more than that of neopentane.

Melting point depends upon packing in crystal lattice whereas boiling point depends upon the surface area of the molecule.

85. (3)

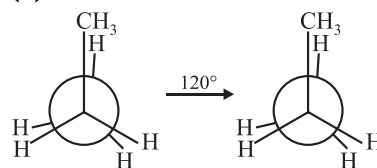
Metal reacts with acid to produce hydrogen gas. Therefore, iron reacts with hydrogen chloride gives hydrogen gas.

In electrochemical series, hydrogen is better reducing agent than iron, therefore when dilute hydrochloric acid is added to iron filings, iron(II) chloride then hydrogen gas is produced. In this reaction iron displaces hydrogen from hydrochloric acid to form iron chloride & hydrogen gas this reaction is a single displacement reaction.

86. (1)

$\text{Cr}_2\text{O}_7^{2-}$	+6
MnO_4^-	+7
VO_3^-	+5
FeF_6^{3-}	+3

87. (3)



If the H-atom eclipsed by the methyl group is held stationary and rotate the front carbon through the sigma bond, after two rotation (each of 60°), the same conformation is observed.

88. (1)

Oxidising agent	Decreases its oxidation number
Reducing agent	Increases its oxidation number
$2\text{Cu}^+ \rightarrow \text{Cu}^{2+} + \text{Cu}^0$	Disproportionation reaction
$\text{H}_2\text{O}_2 + \text{O}_3 \rightarrow \text{H}_2\text{O} + 2\text{O}_2$	Redox reaction

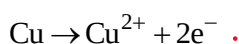
89. (1)

n-Alkanes cannot be easily oxidised to corresponding alcohols with KMnO_4 . Ordinarily alkanes resist oxidation but alkanes having tertiary H-atom can be oxidized to corresponding alcohols by potassium permanganate.

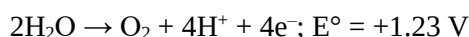
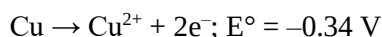
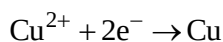
90. (3)

Copper is dissolved at the anode and deposited at the cathode when Cu electrodes are used and the electrolyte is 1 M $\text{CuSO}_4(\text{aq})$ solution. The solution is in the standard concentration so the oxidation potential of Cu is lower than the standard oxidation potential of water.

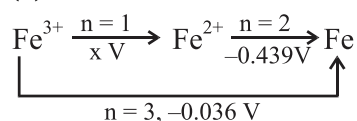
At anode:



At cathode



91. (2)

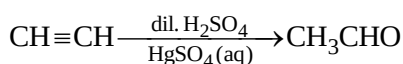


$$-3F(-0.036) = -1F(x) - 2F(-0.439)$$

$$0.108 = -x + 0.878$$

$$x = 0.878 - 0.108 = 0.770 \text{ V}$$

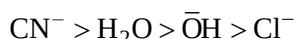
92. (3)



93. (3)

The complex with greater field strength of ligand will be most stable.

The decreasing order of the ligand field strength is:



94. (2)

$$c = 10^{-4} \text{ mol/cm}^3$$

Resistance $R = 50 \text{ ohm}$

$$\text{Conductance } G = \frac{1}{R} = \frac{1}{50} = 0.02 \text{ ohm}^{-1}$$

$$\text{The cell constant is } G^* = \frac{l}{A} = \frac{0.02}{0.004}$$

The specific conductance is $\kappa = GG^*$

Substituting the values, we get

$$\kappa = \frac{1}{50} \times \frac{0.02}{0.004}$$

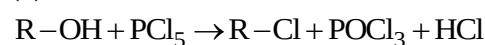
$$\kappa = 0.1 \frac{\text{S}}{\text{m}} = 10^{-3} \text{ S/cm}$$

$$\text{The molar conductance } \lambda_m = \frac{\kappa}{c} = \frac{10^{-3}}{10^{-4}} = 10 \text{ S cm}^2 \text{ mol}^{-1}$$

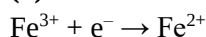
95. (4)

Cu^{2+} ions get reduced easily than H^+ ion. Cu^{2+} ions get reduced easily than H^+ ion. Hydrogen ions cannot oxidise Cu. Copper does not dissolve in HCl.

96. (2)



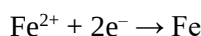
97. (1)



1 mole of $\text{Fe}^{3+} \rightarrow 1 \text{ F}$ of electricity

0.03 moles of $\text{Fe}^{3+} \Rightarrow 0.03 \text{ F}$ of electricity

Number of Faraday left = $0.72 - 0.03 = 0.042 \text{ F}$



1 mole of $\text{Fe}^{2+} \Rightarrow 2 \text{ F}$ of electricity

or 2 F of electricity electrolysis $\Rightarrow 1 \text{ mole of Fe}^{2+}$

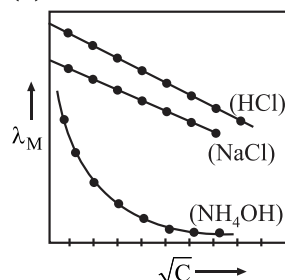
$$0.042 \text{ F of electricity electrolysis} \Rightarrow \frac{1}{2} \times 0.042 = 0.021 \text{ mole of Fe}^{2+}$$

Number of moles of Fe^{2+} unelectrolysed = $0.03 - 0.021 = 0.009 \text{ moles}$

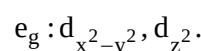
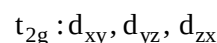
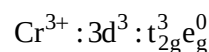
98. (3)

Some coordination compounds of platinum effectively inhibit the growth of tumours. Examples are: cis-platin and related compounds.

99. (2)



100. (3)



SECTION-(III) BOTANY

101. (2)

The exponential growth can be expressed as

$$W_1 = W_0 e^{rt}$$

W_1 = final size (weight, height, number etc.)

W_0 = initial size at the beginning of the period

r = growth rate

t = time of growth

e = base of natural logarithms

[New NCERT Class 11th Page No. 170]

102. (3)

In an isobilateral leaf, the stomata are present on both the surfaces of the epidermis; and the mesophyll is not differentiated into palisade and spongy parenchyma.

[New NCERT Class 11th Page No. 77]

103. (3)



It is showing perigynous condition in rose.

[New NCERT Class 11th Page No. 62]

104. (2)

Meristematic cell possess large conspicuous nucleus.

[New NCERT Class 11th Page No. 168]

105. (1)

- Auxin is called plant growth promoter.
- Auxins promote the abscission of older mature leaves and fruits.

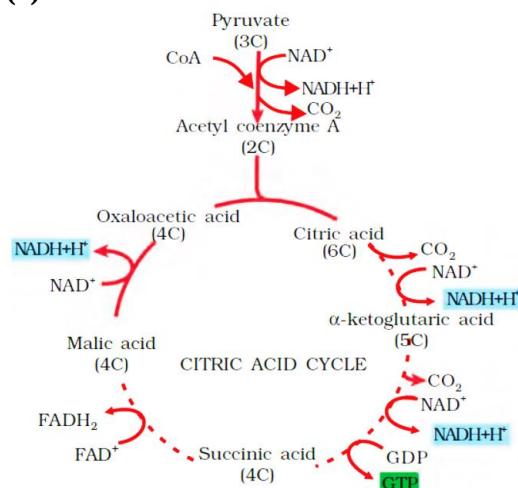
[New NCERT Class 11th Page No. 174, 175]

106. (1)

The 'bakane' (foolish seedling) disease of rice seedlings, was caused by a fungal pathogen *Gibberella fujikuroi*. E. Kurosawa in 1926 reported the appearance of symptoms of the disease in uninfected rice seedlings when they were treated with sterile filtrates of the fungus. The active substances were later identified as gibberellic acid.

[New NCERT Class 11th Page No. 176]

107. (4)



[New NCERT Class 11th Page No. 159]

108. (2)

In some plants, like grass, *Monstera* and the banyan tree, roots arise from parts of the plant other than the radicle and are called adventitious roots.

[New NCERT Class 11th Page No. 58]

109. (4)

In parietal placentation, the ovules develop on the inner wall of the ovary or on peripheral part. Ovary is one-chambered but it becomes two chambered due to the formation of the false septum, e.g., mustard and *Argemone*.

[New NCERT Class 11th Page No. 65]

110. (3)

Cuticle	Waxy layer
Trichomes	Usually multicellular
Root hair	Unicellular
Stomata	Pore

[New NCERT Class 11th Page No. 72]

111. (2)

The biomolecule which is common to a respiration-mediated breakdown of fats, carbohydrates and proteins is acetyl CoA.

[New NCERT Class 11th Page No. 163]

112. (4)

In some leguminous plants the leafbase may become swollen, which is called the pulvinus.

[New NCERT Class 11th Page No. 60]

113. (1)

The outermost layer is epiblema. Many of the cells of epiblema protrude in the form of unicellular root hairs.

[New NCERT Class 11th Page No. 73]

114. (1)

Cuticle and mesophyll are absent in roots.

[New NCERT Class 11th Page No. 72]

115. (2)

- Bundle sheath – RuBisCo
- Bundle sheath – Decarboxylation
- Mesophyll cell – PEP carboxylase
- Oxalo acetic acid – Mesophyll cell

[New NCERT Class 11th Page No. 145]

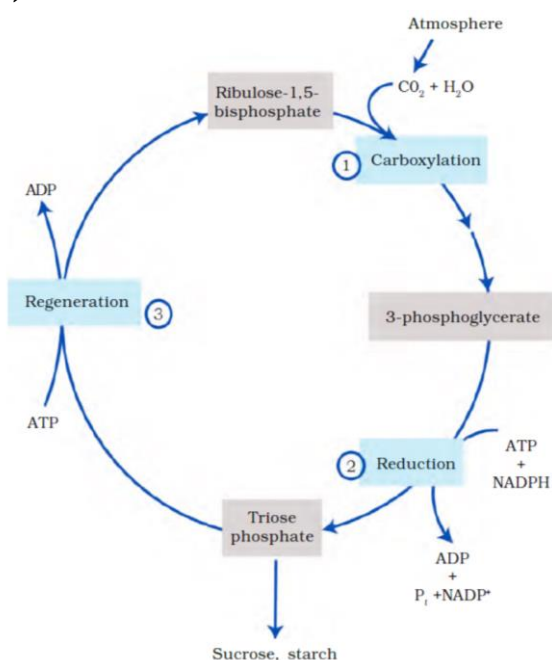
116. (2)

Polyarch xylem – Monocot root

As compared to the dicot root which have fewer xylem bundles, there are usually more than six (polyarch) xylem bundles in the monocot root.

[New NCERT Class 11th Page No. 74]

117. (2)



The first stable product of Calvin cycle has 3 carbon atoms and 1 phosphate group.

[New NCERT Class 11th Page No. 144]

118. (2)

Each flower normally has four floral whorls, viz., calyx, corolla, androecium and gynoecium.

[New NCERT Class 11th Page No. 62, 63]

119. (4)



A-Valvate, B-Twisted and C-Imbricate

[New NCERT Class 11th Page No. 64]

120. (3)

Epidermal cells are parenchymatous with a small amount of cytoplasm lining the cell wall and a large vacuole.

[New NCERT Class 11th Page No. 72]

121. (4)

Plants follow different pathways in response to environment or phases of life to form different kinds of structures. This ability is called plasticity, e.g., heterophylly in cotton, coriander and larkspur. In such plants, the leaves of the juvenile plant are different in shape from those in mature plants. On the other hand, difference in shapes of leaves produced in air and those produced in water in buttercup also represent the heterophyllous development due to environment. This phenomenon of heterophylly is an example of plasticity.

[New NCERT Class 11th Page No. 173]

122. (2)

The fruit is a characteristic feature of the flowering plants. It is a mature or ripened ovary, developed after fertilisation. Generally, the fruit consists of a wall or pericarp and seeds. The pericarp may be dry or fleshy. When pericarp is thick and fleshy, it is differentiated into the outer epicarp, the middle mesocarp and the inner endocarp. In mango and coconut, the fruit is known as drupe.

[New NCERT Class 11th Page No. 65]

123. (4)

On the basis of structure and location, there are three types of tissue systems. These are the epidermal tissue system, the ground or fundamental tissue system and the vascular or conducting tissue system.

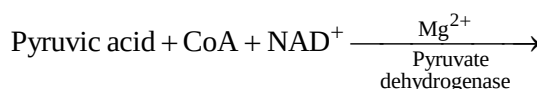
[New NCERT Class 11th Page No. 71]

124. (2)

The ratio of the volume of CO₂ evolved to the volume of O₂ consumed in respiration is called the respiratory quotient (RQ) or respiratory ratio.

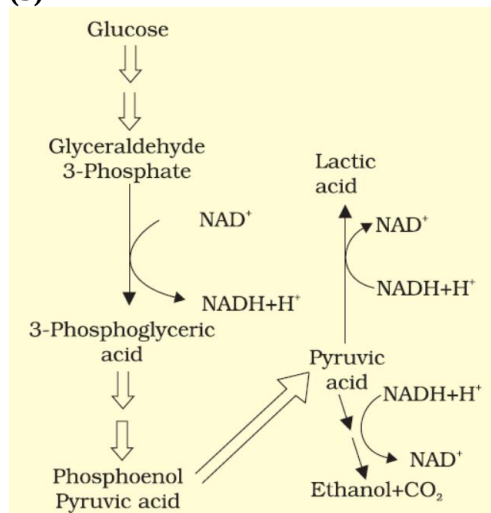
[New NCERT Class 11th Page No. 163]

125. (4)



[New NCERT Class 11th Page No. 158]

126. (3)



[New NCERT Class 11th Page No. 157]

127. (1)

Monoadelphous	China rose
Polyadelphous	Citrus
Diadelphous	Pea
Epipetalous	Brinjal

[New NCERT Class 11th Page No. 64]

128. (1)

GA₃ is the most intensively studied form.

[New NCERT Class 11th Page No. 176]

129. (2)

Bracteate in flower is reduced leaf found at the base of the pedicel.

[New NCERT Class 11th Page No. 62]

130. (1)

The F₁ headpiece is a peripheral membrane protein complex and contains the site for synthesis of ATP from ADP and inorganic phosphate. F₀ is an integral membrane protein complex that forms the channel through which protons cross the inner membrane.

[New NCERT Class 11th Page No. 161]

131. (1)

The C₄ plants show saturation at about 360 μL^{-1} while C₃ responds to increased CO₂ concentration and saturation is seen only beyond 450 μL^{-1} .

[New NCERT Class 11th Page No. 150]

132. (2)

A part of a leaf is enclosed in a test tube containing some KOH soaked cotton (which absorbs CO₂), while the other half is exposed to air. The setup is then placed in light for some time. On testing for the presence of starch later in the two parts of the leaf, you must have found that the exposed part of the leaf tested positive for starch while the portion that was in the tube, tested negative. This showed that CO₂ was required for photosynthesis.

[New NCERT Class 11th Page No. 134]

133. (3)

18 ATP are used to form one hexose molecule.

Stroma lamellae membranes lack PS II as well as NADP reductase enzyme.

[New NCERT Class 11th Page No. 140]

134. (4)

The cells arranged in multiple layers between epidermis and pericycle constitute the cortex. It consists of three sub-zones. The outer hypodermis, consists of a few layers of collenchymatous cells just below the epidermis, which provide mechanical strength to the young stem. Cortical layers below hypodermis consist of rounded thin walled parenchymatous cells with conspicuous intercellular spaces. The innermost layer of the cortex is called the endodermis.

[New NCERT Class 11th Page No. 75]

135. (2)

The LHC are made up of hundreds of pigment molecules bound to proteins.

[New NCERT Class 11th Page No. 138]

136. (4)

In the photorespiratory pathway there is no synthesis of ATP or NADPH and sugar is also not synthesised.

[New NCERT Class 11th Page No. 147]

137. (1)

RuBisCO has a much greater affinity for CO₂ when the CO₂ : O₂ is nearly equal. Imagine what would happen if this were not so This binding is competitive.

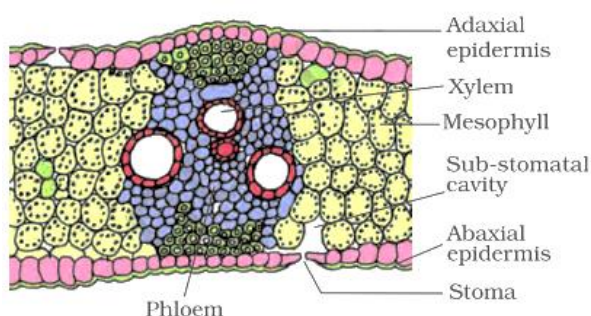
[New NCERT Class 11th Page No. 147]

138. (4)

Oxidative decarboxylation	Pyruvate dehydrogenase
Glycolysis	EMP pathway
Oxidative phosphorylation	Electron transport system
Tricarboxylic acid cycle	Citrate synthase

[New NCERT Class 11th Page No. 155, 158, 159]

139. (3)



[New NCERT Class 11th Page No. 76]

140. (3)

Formation of NADPH is associated with photosystem-I.

[New NCERT Class 11th Page No. 76]

141. (2)

There is a linear relationship between incident light and CO₂ fixation rates at low light intensities.

[New NCERT Class 11th Page No. 149]

142. (4)

- A. Chlorophyll *a* - Bright or blue green
- B. Chlorophyll *b* - Yellow green
- C. Xanthophylls - Yellow
- D. Carotenoids - Yellow to yellow-orange

[New NCERT Class 11th Page No. 137]

143. (2)



[New NCERT Class 11th Page No. 155]

144. (3)

- Glycolysis can use fructose also as substrate.

- Glucose and fructose are phosphorylated to give rise to glucose-6-phosphate by the activity of the enzyme hexokinase.

[New NCERT Class 11th Page No. 156]

145. (1)

Glyceraldehyde-3-phosphate to (1,3 bisphosphoglyceric acid conversion yield NADH + H⁺.

[New NCERT Class 11th Page No. 155]

146. (2)

Auxin	Controls the xylem differentiation
Abscisic acid	Closure of stomata
Ethylene	Female flowers in cucumbers
Gibberellin	Early seed production in conifers

[New NCERT Class 11th Page No. 176, 177]

147. (1)

T.W Engelmann using a prism split light into its spectral components.

The bacteria were used to detect the sites of O₂ evolution.

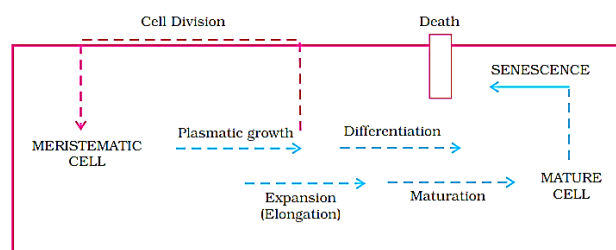
[New NCERT Class 11th Page No. 135]

148. (4)

The enzymes of dark reaction in C₄ plants are found in the stroma of chloroplast of mesophyll cells and bundle sheath cells.

[New NCERT Class 11th Page No. 146]

149. (4)



[New NCERT Class 11th Page No. 173]

150. (1)

Relative growth rate which is also referred to as efficiency index also the measure of the ability of the plant to produce new plant material.

[New NCERT Class 11th Page No. 171]

SECTION-(IV) ZOOLOGY**151. (4)**

Hindbrain comprises pons, medulla and cerebellum.

[New NCERT Class 11th Page No. 236]

152. (4)

Number of nephrons, length of loop of Henle and presence of vasa recta can be considered as point of difference between cortical and juxtamedullary nephrons.

[New NCERT Class 11th Page No. 208]

153. (1)

- In amphibians and reptiles, the left atrium receives oxygenated blood from the gills/lungs/skin and the right atrium gets the deoxygenated blood from other body parts.
- They get mixed up in the single ventricle which pumps out mixed blood (incomplete double circulation).

[New NCERT Class 11th Page No. 197]

154. (2)

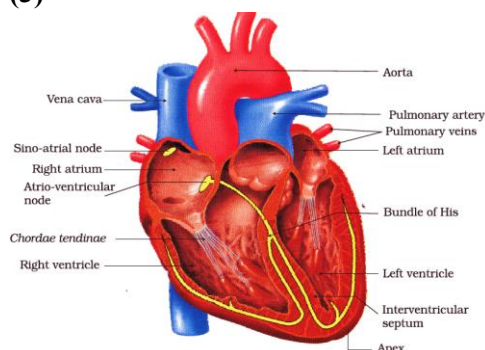
Each actin (thin) filament is made of two 'F' (filamentous) actins helically wound to each other.

[New NCERT Class 11th Page No. 221]

155. (1)

Depolarisation of the axonal membrane is caused due to rapid influx of Na^+ ions when a stimulus is applied at a site.

[New NCERT Class 11th Page No. 232]

156. (3)

III-Bundle of His transmits action potential to the ventricles.

[New NCERT Class 11th Page No. 198,199]

157. (2)

Gastric inhibitory peptide inhibits gastric secretion and motility.

[New NCERT Class 11th Page no. 247]

158. (1)

- The albumins help in osmotic balance of the body.
- Globulins primarily are involved in defense mechanisms of the body.
- Fibrinogens are needed for clotting or coagulation of blood.

[New NCERT Class 11th Page No. 194]

159. (1)

Ovary is the primary female sex organ which produces one ovum during each menstrual cycle.

[New NCERT Class 11th Page No. 246]

160. (1)

- Non-endocrine tissues secrete hormones called growth factors.
- These factors are essential for the normal growth of tissues and their repairing/regeneration.

[New NCERT Class 11th Page No. 247]

161. (3)

- Each branch terminates as a bulb-like structure called synaptic knob which possess synaptic vesicles containing neurotransmitters in it.
- Multipolar neurons are found in the cerebral cortex of the brain.

[New NCERT Class 11th Page No. 232]

162. (3)

Melanocyte stimulating hormone	Pars intermedia
Thyroid stimulating hormone	Pars distalis
Thyroxine	Thyroid gland
Corticoids	Adrenal gland

[New NCERT Class 11th Page No. 241-244]

163. (4)

Parathyroid hormone (PTH) increases the Ca^{2+} levels in the blood and is a hypercalcemic hormone.

[New NCERT Class 11th Page No. 243]

164. (2)

- Lub is the first heart sound which is associated with the closure of the tricuspid and bicuspid valves.
- Dub is the second heart sound which is associated with the closure of the semilunar valves.

[New NCERT Class 11th Page No. 200]

165. (2)

- The nodes of Ranvier are the interruptions between myelin sheaths on the axon of myelinated nerve fibres.
- They help in impulse conduction across the neuron in a faster manner.

[New NCERT Class 11th Page No. 232]

166. (3)

An impairment affecting synthesis or release of antidiuretic hormone (ADH) results in a diminished ability of the kidney to conserve water leading to water loss and dehydration. This condition is known as Diabetes Insipidus.

[New NCERT Class 11th Page No. 242]

167. (3)

- By releasing the ADP and P_i, the myosin goes back to its relaxed state.
- Repeated activation of the skeletal muscles can lead to the accumulation of lactic acid.
- Skeletal muscle contraction is initiated by a signal sent by the central nervous system (CNS).
- The skeletal muscle bundles are held together by a common collagenous connective tissue layer called fascia.
- Muscles are mesodermally derived.

[New NCERT Class 11th Page no. 218, 219, 222]

168. (4)

Carpals (wrist bones) are 8 in number while tarsals (ankle bones) are 7 in number.

[New NCERT Class 11th Page no. 226]

169. (4)

Electrical synapse	Rare in human system
Afferent nerve fibres	Transmit impulses from tissues/ organs to the CNS
Bipolar neurons	Retina of eyes
Axon	Transmit impulses away from the cell body

[New NCERT Class 11th Page No. 232,234]

170. (4)

The protein, peptide hormones and amino acid derived hormones interact with membrane-bound receptors.

- Epinephrine is an amino acid derivative which binds to its receptors present on the plasma membrane.
- Follicle stimulating hormone (FSH) is a polypeptide hormone which binds to its receptors present on the plasma membrane.
- Steroid hormones and iodothyronines interact with intracellular receptors and mostly regulate gene expression or chromosome function by the interaction of hormone-receptor complex with the genome.

[New NCERT Class 11th Page No. 248]

171. (4)

Eosinophils	Allergic reactions
Basophils	Secrete histamines, heparin and serotonin and are involved in Inflammatory reactions
Monocytes	Phagocytic action
Neutrophils	Phagocytic action

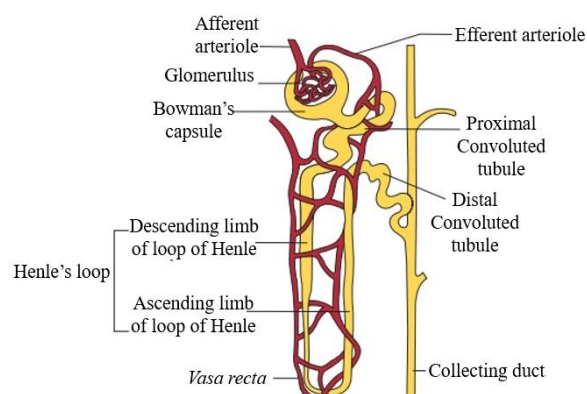
[New NCERT Class 11th Page No. 194,195]

172. (3)

Flatworm	Flame cell
Prawn	Green gland
<i>Periplaneta</i>	Malpighian tubule
Mammals	Kidney

[New NCERT Class 11th Page No. 206]

173. (4)



In collecting duct of nephron, the filtrate is called urine.

[New NCERT Class 11th Page No. 207]

174. (4)

In response, the stretch receptors on the walls of the bladder send signals to the CNS. The CNS passes on motor messages to initiate the contraction of smooth muscles of the bladder and simultaneous relaxation of the urethral sphincter causing the release of urine.

[New NCERT Class 11th Page No. 212,213]

175. (2)

- Bowman's capsule is double walled cup-like structure.
- Glomerulus along with Bowman's capsule, is called the Malpighian body or renal corpuscle.
- Elimination of uric acid requires the least amount of water.
- Urine formation involves filtration, reabsorption and secretion.

[New NCERT Class 11th Page No. 205,206,208]

176. (2)

Somatostatin from the hypothalamus inhibits the release of growth hormone from the anterior pituitary gland.

[New NCERT Class 11th Page No. 240]

177. (1)

During cardiac cycle the flow of blood into the ventricles increases by about 30 per cent during atrial systole.

[New NCERT Class 11th Page No. 199]

178. (4)

- The two main types of cells in the Islet of Langerhans of pancreas are called α -cells and β -cells.
- The α -cells secrete a hormone called glucagon, while the β -cells secrete insulin.

[New NCERT Class 11th Page No. 245]

179. (3)

Glomerulonephritis is the inflammation of the glomeruli.

[New NCERT Class 11th Page No. 214]

180. (4)

On an average, 1100 – 1200 ml of blood is filtered by the kidneys per minute which constitute roughly $1/5^{\text{th}}$ of the blood pumped out by each ventricle of the heart in a minute.

[New NCERT Class 11th Page No. 208]

181. (4)

Hinge joint	–	Knee joint
Pivot joint	–	Between the carpals
Cartilaginous joint	–	Between the adjacent vertebrae in the vertebral column
Fibrous joint	–	Flat skull bones

[New NCERT Class 11th Page no. 227]

182. (4)

All vertebrates (Fishes, respites, amphibians and birds) have a closed circulatory system

[New NCERT Class 11th Page No. 197]

183. (1)

During the shortening of the skeletal muscle, i.e., contraction, the 'I' bands get reduced, whereas the 'A' bands retain the length and the H zone disappears.

[New NCERT Class 11th Page no. 222]

184. (1)

- Tropomyosin runs along the length of the actin filament, interacting with the F-actin filaments.
- Myosin is made up of monomeric proteins called meromyosins.

[New NCERT Class 11th Page no. 221,222]

185. (4)

- Our vertebral column is formed by 26 serially arranged units called vertebrae and is dorsally placed.
- Smooth muscles activities are not under the voluntary control of the nervous system.
- H zone is the central part of thick filament, not overlapped by thin filaments.
- In the centre of each 'I' band is an elastic fibre called the 'Z' line which bisects it.
- There are two pairs of floating ribs in the human body.

[New NCERT Class 11th Page no. 218, 219, 220,225]

186. (1)

Sternum	One
Femur	Two
Metatarsals	Ten
Cranial bones	Eight

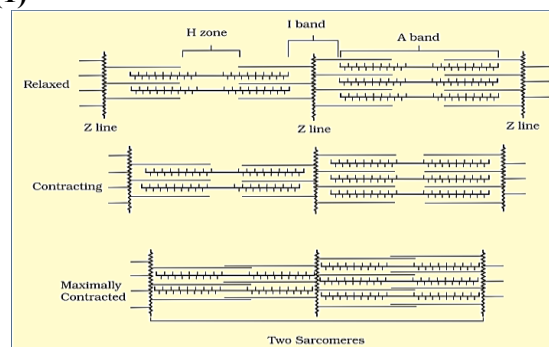
[New NCERT Class 11th Page no. 225, 226]

187. (4)

- The association areas are responsible for complex functions like intersensory associations, memory and communication.
- Body temperature is regulated by the centers present in the hypothalamus.
- Expression of emotional reactions is the function of the limbic system.

[New NCERT Class 11th Page No. 236]

188. (1)



- In the given diagram 'I' represents relaxed state of muscle.
- The Ca^{++} ions are pumped back to the sarcoplasmic cisternae resulting in the unmasking of actin filaments, leads to the return of 'Z' lines back to their original position, i.e., relaxation.

[New NCERT Class 11th Page no. 223]

189. (4)

Steroids hormones and iodothyronines can easily pass through the cell membrane of target cell and bind to specific intracellular receptors.

[New NCERT Class 11th Page no. 248]

190. (3)

- Thyroxines play an important role in the regulation of the basal metabolic rate.
- Melatonin plays a very important role in the regulation of a 24-hour (diurnal) rhythm of our body.

[New NCERT Class 11th Page No. 242,243]

191. (1)

The first vertebra is the atlas and it articulates with the occipital condyles and forms a dicondylic skull.

[New NCERT Class 11th Page no. 225]

192. (3)

- Atrial natriuretic factor (ANF) can cause vasodilation and thereby decrease the blood pressure
- Conditional reabsorption of Na⁺ and water takes place in the distal convoluted tubule of the renal tubule.

[New NCERT Class 11th Page No. 212,209]

193. (2)

The pars distalis region of pituitary gland, commonly called anterior pituitary, produces growth hormone (GH), prolactin (PRL), thyroid stimulating hormone, (TSH), adrenocorticotrophic hormone (ACTH), luteinizing hormone (LH) and follicle stimulating hormone (FSH).

[New NCERT Class 11th Page No. 241]

194. (3)

The axonal membrane is comparatively more permeable to potassium ions and nearly impermeable to sodium ions during resting state.

[New NCERT Class 11th Page No. 233]

195. (4)

- Human kidneys can produce urine nearly four times concentrated than the initial filtrate formed.
- Reabsorption of water occurs passively in the initial segments of the nephron.

[New NCERT Class 11th Page No. 212,209]

196. (4)

Angiotensin-II increase reabsorption of Na⁺ from the distal part of renal tubules.

[New NCERT Class 11th Page No. 212]

197. (4)

Tetany	Rapid spasms (wild contractions) in muscle due to low Ca ⁺⁺ in body fluid.
Myasthenia gravis	An autoimmune disorder affecting the neuromuscular junction leading to fatigue, weakening and paralysis of skeletal muscle.
Muscular dystrophy	Progressive degeneration of skeletal muscle mostly due to genetic disorder.
Arthritis	Inflammation of joints.

[New NCERT Class 11th Page no. 227]

198. (1)

- The myelinated nerve fibres are enveloped with Schwann cells, which form a myelin sheath around the axon.
- Unmyelinated nerve fibre is enclosed by a Schwann cell that does not form a myelin sheath around the axon.

[New NCERT Class 11th Page No. 232]

199. (1)

Protein makes 6-8 percent of plasma which is more to eosinophils (2-3percent) of total leucocyte.

[New NCERT Class 11th Page No. 194]

200. (3)

- Neural signals through the sympathetic nerves (part of ANS) can increase the rate of heart beat.
- Parasympathetic neural signals decrease the rate of heart beat
- Heart is situated in the thoracic cavity, in between the two lungs, slightly tilted to the left.

[New NCERT Class 11th Page No. 198, 202]

